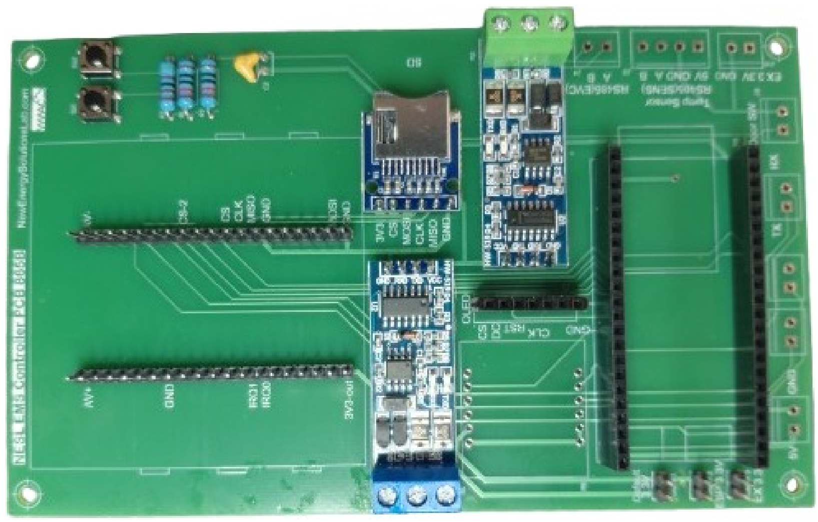
BUILD-EMS

Behind-the-Meter Control for Rooftop Solar, Wind, Storage & EV — Multi-Tenant Ready



Deploying EV charge/discharge with rooftop solar PV and wind on an existing building? **BUILD-EMS** is the open-source behind-the-meter controller that integrates **bidirectional DC/AC inverters**, multi-tenant "plug-in solar", and EV infrastructure into one balanced building minigrid — enforcing multi-zone energy policy while protecting the mains.

What Sets BUILD-EMS Apart:

- **Bidirectional & Inverter-Ready** : coordinates DC/AC bidirectional inverters and converters for solar PV, wind and battery, with EV charge and discharge (V2X).
- **Multi-Tenant "Plug-in Solar"** : partitions shared rooftop generation and storage across tenants with per-zone, time-of-day energy policies.
- **Mains-Protective** : balances the building mains and responds to grid-peak DERMS events without exceeding the service limit.

BUILD-EMS turns a building rooftop into a coordinated, multi-tenant minigrid — maximizing solar, wind and EV flexibility while keeping the mains within limits.

Intelligent

- Multi-zone, time-of-day energy policy
- Mains balancing & DERMS peak response
- EV charge / discharge (V2X) scheduling

Interoperable

- Bidirectional DC/AC inverters (PV / wind / battery)
- SunSpec DER + Modbus / CAN
- Multitenant sub-metering & EVSE

Resilient

- Local policy enforcement, cloud-optional
- Per-tenant energy allocation
- Open meshEMS platform

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Product Specifications

PRODUCT	
Product name	BUILD-EMS (10Power X-EMS, supplied by NESL)
Model / PCB	NESL EMS Controller PCB 886B (shared X-EMS platform)
Firmware	meshEMS / OpenAMI metering (open source, ∼50k LOC middleware)
COMPUTE	
Processor	ESP32-S3-WROOM-1 | dual-core Xtensa LX7 @ 240 MHz
Memory	16 MB Flash + 8 MB PSRAM
ENERGY MANAGEMENT	
Supported DER	Solar PV | wind | battery, via bidirectional DC/AC inverters / converters
EV infrastructure	EVSE charge & discharge (V2X) | per-tenant charging needs
Energy policy	Multi-zone time-of-day policies | per-channel kW / kWh limits | multitenant allocation | DERMS response
ASSET COMMUNICATION	
Serial buses	2× RS-485 Modbus: EV-charger (EVC) + sensor/DTM ports | 1× CAN 2.0
On-board metering	SPI poly-phase metering, up to 6 CT channels (CircuitSetup ATM90E32) | bidirectional (import/export)
Local I/O	2× onboard SSR outputs | door-switch input | 2 analog buttons
SUPPORTED MODULES & SENSORS (out of the box)	
DER & inverters	SunSpec models 1 / 11 / 213 (bidirectional inverters & meters)
Energy meters / relay	DDS238 & CHINT CHD130 (bidirectional kWh) | 8-ch I^2C SSR bank (PCF8574, *latest*)
Sensors / networking	SHT20 (temp / humidity) | Ethernet | WiFi / BLE
POWER & ENCLOSURE	
Operating voltage	5 V DC — USB-C or screw terminals *[to confirm]*
Power / dimensions	*[to confirm]*
STANDARDS & POSITIONING	
Grid position	Behind-the-meter (BTM) | building / commercial rooftop minigrid
Capabilities	EV V2X | multitenant solar / wind / storage | mains balancing & DERMS

Customizable with added hardware modules and open or licensed code stacks. Specifications are preliminary; items marked [to confirm] are to be confirmed on the production unit. BUILD-EMS is an open-source X-EMS product supplied by New Energy Solutions Lab for 10Power.